

Marcus Ma

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EDUCATION

University of Southern California

PhD in Computer Science (SAIL, Advisor: Shrikanth Narayanan)

August 2024 – Present

- 4.0 GPA: Courses taken: PSYC 626 Text as Data, CSCI 670 Advanced Analysis of Algorithms, CSCI 699 History of Language and Computing

Georgia Institute of Technology

MS in Computer Science, Machine Learning Specialization (Advisor: Wei Xu)

August 2023 – May 2024

- 4.0 GPA; Relevant Coursework: Game AI, Advanced Natural Language Processing, Theory of ML
- Master's Thesis: leveraging novel NLP techniques and LLMs for cross-genre authorship attribution and verification; creating a new SOTA domain-adaptive attribution model for multi-genre documents

BS in Computer Science, Intelligence and Theory Concentration

August 2020 – May 2023

- 4.0 GPA (graduated summa cum laude, Dean's List every semester)

HONORS & AWARDS

- Andrew and Erna Viterbi Graduate Student Fellowship, USC (2024-2026)
- DAAD RISE Research Scholarship, German Academic Exchange Service (2022)

SELECTED PUBLICATIONS

- **Ma, M.**, Narayanan, S. Paired Multimodal Scaling Laws. Under Review at ICLR 2027.
- **Ma, M.**, Narayanan, S. Intelligence Requires Grounding Beyond Embodiment. Under Review at AAAI 2027.
- **Ma, M.**, Zhou, E., Ludwig, L., Hörath, J., Winkler, C., Avramidis, K., Feng, T., Toth, G., Bothe, A., Narayanan, S. Smiling Regulates Emotion During Traumatic Recollection. Proceedings of the 28th ACM International Conference on Multimodal Interaction (ICMI), 2026.
- Zhou, E., **Ma, M.**, Avramidis, K., Toth, G. M., Narayanan, S. Looking Into the Past: Eye Movements Characterize Elements of Autobiographical Recall in Interviews with Holocaust Survivors. arXiv preprint arXiv:2604.22016, Apr 2026.
- **Ma, M.**, Prescott, J., Zhou, E., Feng, T., Avramidis, K., Toth, G., Narayanan, S. Encoding Emotion Through Self-Supervised Eye Movement Reconstruction. Proceedings of ICASSP 2026, May 2026.
- Avramidis, K., Jeong, W., Kommineni, A., Kadiri, S., **Ma, M.**, McDaniel, C., Hughes, M., McGee, T., Kaiser, E., Byrd, D., Habibi, A., Cahn, B., Blank, I., Lerman, K., Medani, T., Leahy, R., Narayanan, S. Deep Learning Characterizes Depression and Suicidal Ideation in Young Adults from Eye Movements. npj Digital Medicine, 9(1), Mar 2026.
- **Ma, M.**, Narayanan, S. Intelligence Requires Grounding But Not Embodiment. arXiv preprint arXiv:2601.17588, Jan 2026.
- **Ma, M.**, Johnson, C., Bridges, N., Trager, J., Chochlakis, G., Narayanan, S. Authors Should Label Their Own Documents. arXiv preprint arXiv:2512.12976, Dec 2025.
- **Ma, M.**, Chochlakis, G., Pandiyan, N. M., Thomason, J., Narayanan, S. Large Language Models Do Multi-Label Classification Differently. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Nov 2025.
- Chochlakis, G., Wu, P., Bedi, T., **Ma, M.**, Lerman, K., Narayanan, S. Humans Hallucinate Too: Language Models Identify and Correct Subjective Annotation Errors With Label-in-a-Haystack Prompts. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Nov 2025.
- Hall, K., **Ma, M.**, Zhang, X., Das Swain, V., Kim, Jennifer. Understanding Human-AI Misalignment in LLM-Based Job-Seeking Support for Neurodivergent Users. Proceedings of the 27th International ACM SIGACCESS Conference on Computers and Accessibility, Oct 2025.

- Shi, X., Zhang, Y., Lu, Y., **Ma, M.**, Feng, T., Toutios, A., Hsu, H., Goldstein, L., Narayanan, S. 75-Speaker Annot-16: A benchmark dataset for speech articulatory rt-MRI annotation with articulator contours and phonetic alignment. Interspeech 2025.
- Toth, G. M., Laib, M., Bothe, A., **Ma, M.**, Narayanan, S., Pruski, C., Silveria, M. “Crying like a Baby”: Survival Analysis and the Multimodal Memory of Holocaust Survivors. Anthology of Computers and the Humanities, Apr 2025.
- **Ma, M.**, Le, D., Kang, J., Dou, Y., Cadigan, J., Freitag, D., Ritter, A., Xu, W. CROSSNEWS: A Cross-Genre Authorship Verification and Attribution Benchmark. Proceedings of the AAAI Conference on Artificial Intelligence, 39(23), 2025.
- Seeburger, D. T., Xu, N., **Ma, M.**, Larson, S., Godwin, C., Keilholz, S. D., Schumacher, E. H. Time-Varying Functional Connectivity Predicts Fluctuations in Sustained Attention in a Serial Tapping Task. Cognitive, Affective, & Behavioral Neuroscience, 24(1), Jan 2024.
- **Ma, M.**, Glass, L. and Stanford, J. Introducing Bed Word: A New Automated Speech Recognition Tool for Sociolinguistic Interview Transcription. Linguistics Vanguard, 2024.
- **Ma, M.**, Kim, C., Hall, K., Kim, J. It Takes Two to Avoid Pregnancy: Addressing Conflicting Perceptions of Birth Control Pill Responsibility in Romantic Relationships. Proc. ACM Human-Computer Interaction. 7, CSCW2, Article 282 (October 2023).
- **Ma, M.** and Glass, L. (2022, Oct 13-15). BedWord: Computer-generated, human-edited TextGrids for faster sociolinguistic transcription. NWAV 50 Conference, Palo Alto, CA.
- Seeburger, D., Xu, N., Godwin, C., **Ma, M.**, Keilholz, S., Schumacher, E. (2022, Apr 24-27). Identifying the Neural Mechanisms of Zone State Performance using Time-varying Functional Connectivity Methods. CNS 2022 Conference, San Francisco, CA.

ONGOING RESEARCH PROJECTS

Environmental Factors of General Intelligence

2025 – Present

USC SAIL (Narayanan)

- Argued in a position paper that intelligence requires grounding (consistent, causal assignment of meaning to symbols through interaction with an environment) but not embodiment (Under Review at AAAI 2027)
- Introduced a four-channel multimodal scaling law based on partial information decomposition, showing that only paired data reduces synergistic loss and that synergy emerges only past a critical number of pairs (Under Review at ICLR 2027)

Nonverbal Emotion and Cognitive State Analysis

2025 – Present

USC SAIL (Narayanan) + USC Shoah Foundation

- Pretrained a self-supervised gaze model on thousands of hours of unlabeled, low-resolution Holocaust survivor testimonies to predict speech-derived valence, arousal, and dominance and behaviors such as laughing and crying (ICASSP 2026)
- Built an automatic smile detector (85% F1) and aligned smiles from 978 survivors with narrative and multimodal valence features, finding negative-affect smiles precede more positive valence trajectories (ICMI 2026)
- Analyzing how eye movements characterize episodic, semantic, affective, and temporal dimensions of autobiographical recall across 806 interviews

LLM Subjectivity Bias (Published in EMNLP 2025)

2024 – 2025

USC SAIL (Narayanan)

- Investigating prior biases of LLMs by investigating how often they attend to incorrect information directly provided during in-context learning versus relying on their pretrained internal representations
- Explaining multi-label, multi-class probability distributions through an aggregation of binary choice prompts rather than a single multi-label prompt distribution which inaccurately reflects model beliefs

LLM-Brain Alignment

2024 – 2025

USC SAIL (Narayanan) + Morality and Language Lab (Dehghani)

- Correlating internal LLM representations of language stimuli to fMRI activations to determine how and where brains and language models reason similarly

- Writing a review paper comparing whether the field of LLM-Brain alignment provides supporting evidence that humans and LLMs align in hierarchical predictive coding for language production and understanding

Automatic Vocal Tract Segmentation (Published in Interspeech 2025)

2024 – 2025

USC SAIL (Narayanan) + SPAN

- Created a fully-automated pipeline for vocal tract area segmentation using a combination of hand-annotations and finetuned SAM-2 object segmentation models, increasing accuracy compared to off-the-shelf SAM2 by 70% and other vocal tract segmentation models by 15%
- Currently creating an MRI-to-EMG synthesis model for downstream use for language production from silent mid-sagittal rt-MRI videos

Adolescent Suicidality

2024 – Present

USC SAIL (Narayanan) + Children's Hospital Los Angeles

- Conducting a retrospective study of emergency department visits by children aged 6–17 at CHLA from 2016–2023 to identify predictors of suicidality and move beyond binary risk assessment
- Deriving NLP features from post-visit clinical notes and comparing model performance with and without them to measure their predictive power for visits and revisits

Exposome Factors for Dementia

2024 – Present

USC SAIL (Narayanan)

- Investigating exposomes (the totality of internal, societal, political, and environmental factors) that influence dementia development as part of a multi-year NIH-led project

TEACHING EXPERIENCE

Graduate Teaching Assistant for CS 4650: Natural Language Processing Jan 2023 – May 2024

- Assist in leading classes of 75+ students through NLP, covering topics from encoders to present-day LLMs
- Hold weekly office hours and grade 200+ assignments per semester

Tutoring and Academic Services: One-on-one Tutor, PLUS Leader

Aug 2021 – Dec 2022

- Tutored over 200 Georgia Tech students in intro-level math and computer science courses
- Prepared and taught ~30 weekly PLUS Sessions (~1200 student cumulative attendance) for Data Structures

PREVIOUS RESEARCH LABS

NLP-X Lab (Natural Language Processing) | Advisors: Wei Xu and Alan Ritter

Cross-Domain Authorship Analysis (Published in AAI 2025)

January 2022 – August 2024

- Created the largest public cross-genre authorship dataset across hundreds of different news sites using a combination of hand-picked gold labels and thousands of silver-labelled authors determined from statistical models of social media network graphs and string linkage methods
- Leveraged this dataset to create a domain-adaptable authorship attribution model trained via contrastive learning on negative-mined paraphrases that reaches SOTA performance on various authorship tasks

Co-Well Lab (Human-Computer Interaction) | Advisor: Jennifer Gahee Kim

Personalized LLMs for Neurodivergent Jobseekers (Published in ASSETS 2025)

Mar 2023 – Aug 2024

- Despite huge gains in conversational AI since the democratization of LLMs, chatbots are often unaccommodating of neurodivergent clients with personalized communication preferences
- Currently leading a pilot study on how to stylize LLMs via prompt engineering to adapt to different strengths and skills in the context of providing career advice to neurodivergent jobseekers
- Analyzed 300+ chat logs and interviewed 15 neurodivergent participants, finding that a GPT-4 career chatbot misrepresented users' skills and imposed neurotypical language and expectations

Contraception Collaboration between Couples (Published in CSCW 2023)

August 2021 – March 2023

- Investigated the expectations placed on women in relationships for taking birth control pills consistently, with findings indicating responsibility lies entirely on women and female-sex individuals

- Developed an app prototype designed to split the responsibility of taking pills more equitably between partners and interviewed nine couples to investigate the intersection of contraception responsibility and technology, findings published in CSCW 2023

Sociolinguistics VIP Team | Advisor: Lelia Glass

Language and Politics in the New South

2021 – 2022

- Served as team leader and ambassador; interviewed 100+ native Georgia residents and analyzed vowel formants, finding a correlation between political conservatism and the Southern Vowel Shift
- Established interview methods, transcription pipelines, and vowel analysis scripts that remain in use by the team

Bed Word: Automated Linguistic Transcription (Published in Linguistics Vanguard 2024)

2022 – 2023

- Built an automatic speech transcription tool that exports to common linguistics analysis formats, now publicly hosted on Dartmouth's DARLA website
- Transcribed ~500 hours of interview audio across 700+ jobs by August 2023, saving an estimated ~1,000 hours of manual transcription labor

CoNTRoL Lab (Cognitive Neuroscience) | Advisor: Eric Schumacher

Neural Zone State Performance (Published in CABN 2024)

2021 – 2022

- Extracted brain-image slices from raw 4D fMRI data and built an algorithm to detect quasi-periodic patterns (QPPs) between the default mode and task positive networks
- Found that QPP activity is more prevalent during in-the-zone periods of focus and attention

SELECTED WORK HISTORY

Echo Group (Valuation: \$25M)

Chief Technology Officer & Co-Founder

2025 – Present

- Lead a team of 3 full-time SWE's and a dozen contractors to build a proprietary chatbot system with 100k US users and 10M+ chats sent that we use to understand population-level trends
- Provide insights to United, NextDoor, Waymo, GSK, J&J, Royal Caribbean, and several ex-Congressmen and political consultants
- Develop and collect real-time, author-labeled preferences directly from human sources rather than LLM-based simulation better, faster, and more accurate than any other alternative on the market

Mercor

LLM Software Engineer: Team Lead | Part-Time

Jan 2025 – July 2025

- Leading team of 150+ professional software engineers in a large-scale LLM human evaluation project for several large AI companies
- Overseeing generation and review of \$500,000+ worth of human-annotated LLM responses for cutting edge NLP research and production model Reinforcement Learning from Human Feedback (RLHF)
- Oversaw collection of expert-annotated math and coding documents used by frontier labs as verifiable rewards for training models such as Gemini, Claude, and GPT Codex

Amazon Web Services

Software Developer Intern: Optical Networking Team | Full-Time Position Offered May 2023 – August 2023

- Spearheaded the transition of optical device networking communication from internal Amazon servers to the more scalable and maintained Native AWS environment, saving ~\$10,000 of yearly server operation costs
- Created a DNS device hostname resolution script that was deployed to hundreds of existing optical devices in the AWS fleet, improving operational efficiency over previous method by ~40%
- Added a robust testing and logging system that monitors the hostname resolution script that automatically initiates rollback in case of failure

Software Developer Intern: Postgres SQL Team | Full-Time Position Offered

May 2022 – August 2022

- Worked in Postgres source code to implement an optimization on the SQL Join command that used semi-join hash filtering to remove unnecessary table rows early in the execution tree
- Improved the speed of certain SQL join commands by up to 36% and integrated filter decisions for the merge join strategy into Postgres query planner, improving AWS Postgres' overall TPC-H database benchmark performance by ~5%

- Created an internal SQL test suite to verify improvement and deployed these tests to the entire AWS codebase, with future work looking to contribute semi-join filtering optimization to an upcoming Postgres community release

Forschungszentrum Jülich (IEK-10)

DAAD RISE Research Intern

Aug 2022 – Oct 2022

- Investigated McCormick relaxations as a layer-propagation technique for verifiable neural network robustness, as an alternative to Interval Bound Propagation (IBP)
- Found McCormick relaxations produce ~40% tighter bounds than IBP, but at a computational cost that grows quadratically with input dimension

Prognomiq

Software Engineer Intern

May 2021 – July 2021

- Built an automated pipeline that converted several thousand heterogeneous medical records from dozens of field studies into a single unified format
- Merged converted records with existing server data and created per-patient version histories for patients with multiple forms

Rocketansky

Full-Stack Engineer (Contract)

Mar 2021 – May 2021

- Developed the main Windows application for a passwordless authentication platform based on hardware keys and biometrics
- Wrote the backend service for per-device cryptographic key generation and authentication handshakes with AWS servers, along with the frontend and Windows installation drivers